

Beyond the Barrier View of Risk: Content Quality, Trust and Informed Adoption in an Extended UTAUT Model of MOOC Acceptance among Educational Personnel

Phanuwat Ruangkulsap¹, Thada Jantakoon¹ & Rukthin Laoha¹

¹Department of Information and Communication Technology for Education, Faculty of Science and Technology, Rajabhat Maha Sarakham University, Maha Sarakham, Thailand

Correspondence: Phanuwat Ruangkulsap, Department of Information and Communication Technology for Education, Faculty of Science and Technology, Rajabhat Maha Sarakham University, Maha Sarakham, Thailand.

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Abstract

Massive open online courses (MOOCs) are increasingly positioned as infrastructure for continuing professional development, yet the mechanisms through which educational personnel accept and use them remain incompletely specified. Drawing on the unified theory of acceptance and use of technology (UTAUT), the information systems success framework, and the trust–risk perspective, this study develops and tests an extended acceptance model that incorporates perceived content quality, trust, perceived risk, self-regulated learning, and digital–AI literacy. We collected survey data from 290 educational personnel in Thailand and screened for insufficient-effort responding, yielding an analytical sample of 270. We estimated the model using partial least squares structural equation modeling with 10,000 bootstrap subsamples, and assessed its predictive capability using PLSpredict and the cross-validated predictive ability test. The model explained 51.7% of the variance in behavioral intention and 42.1% in use behavior. Perceived content quality functioned as the principal upstream driver, exerting a very large effect on self-regulated learning and substantial effects on trust and effort expectancy. Trust was the strongest determinant of behavioral intention and significantly reduced perceived risk. Contrary to the conventional barrier view, perceived risk exerted significant positive effects on both behavioral intention and use behavior, a pattern interpreted as informed adoption. Effort expectancy influenced intention entirely through performance expectancy. This study discusses theoretical and practical implications for platform design and institutional policy.

Keywords: MOOCs, UTAUT, PLS-SEM, trust, perceived risk, self-regulated learning, teacher professional development

1. Introduction

1.1 Introduce the Problem

Massive open online courses (MOOCs) have moved from being an experimental format for widening participation to being treated, in many national systems, as standing infrastructure for the continuing professional development of teachers and other educational personnel. Where earlier deployments framed MOOCs primarily as a means of reaching learners at scale, current policy increasingly positions them as an instrument for upgrading the competencies of the education workforce itself (UNESCO, 2016). This shift changes the analytical problem. Educational personnel are not generic consumers of an online service: they are professionals whose adoption decisions are embedded in institutional mandates, credentialing expectations and their own pedagogical judgments about the quality of the material they are asked to learn from.

Technology acceptance research has supplied a durable vocabulary for such decisions. The unified theory of acceptance and use of technology (UTAUT) posits that performance expectancy, effort expectancy, and social influence shape behavioral intention, while facilitating conditions and intention shape use behavior (Venkatesh, Morris, Davis, & Davis, 2003). Its consumer extension adds hedonic motivation, price value, and habit (Venkatesh, Thong, & Xu, 2012). Applied to MOOCs, these frameworks have repeatedly demonstrated explanatory value (Tseng, Lin, Wang, & Liu, 2022). However, a persistent difficulty is that the core UTAUT constructs describe the perceived properties of a technology in the abstract. In contrast, professional learners

evaluate a specific corpus of instructional content, sustain engagement without external supervision, and weigh whether the resulting credentials will be recognized. This study therefore examines how content quality, trust, perceived risk, and self-regulated learning combine with the UTAUT core to determine MOOC acceptance and use among educational personnel.

1.2 Explore Importance of the Problem

Three considerations make this problem worth new research. First, completion and sustained use remain the practical bottleneck in MOOC-based professional development, and intention-only models cannot address it; a model that ends at behavioral intention leaves the more policy-relevant outcome unmeasured. Second, the treatment of perceived risk in acceptance research has been almost uniformly negative. Risk is modeled as an inhibitor, following the e-commerce tradition in which perceived risk depresses the intention to transact (Featherman & Pavlou, 2003; Pavlou, 2003). Whether that formulation transfers to professional learners who have already accumulated substantial platform experience remains an open empirical question, with direct consequences for how institutions communicate about MOOC credentials. Third, the rapid diffusion of artificial intelligence into learning platforms has made AI literacy a plausible antecedent of technology evaluation among educators (Ng, Leung, Chu, & Qiao, 2021), yet it is rarely modeled jointly with digital literacy, which is empirically difficult to separate.

The purpose of this study is accordingly to specify and test an extended UTAUT model for MOOC acceptance among educational personnel that (a) carries the dependent variable through to self-reported use behavior, (b) positions perceived content quality as an upstream determinant rather than a peripheral correlate, (c) models trust and perceived risk as a linked pair rather than as independent predictors, and (d) evaluates the resulting model not only for explanatory power but for out-of-sample predictive capability.

1.3 Describe Relevant Scholarship

Research on MOOC acceptance has developed along three broadly separate lines. The first applies UTAUT or UTAUT2 more or less directly. Tseng et al. (2022) examined teachers' adoption of MOOCs through a UTAUT2 lens and reported that experience moderates several core relationships, indicating that adoption dynamics differ between novice and experienced users. The second line addresses continuance rather than initial adoption, typically through expectation–confirmation reasoning; Rekha, Shetty and Basri (2023) found that confirmation, perceived usefulness and computer self-efficacy shape satisfaction and continuance among Indian MOOC users, while Shanshan and Wenfei (2024) emphasized psychological stimuli and emotional responses. The third line foregrounds platform quality: Gu, Xu and Sun (2021) demonstrated that MOOC quality attributes influence continuance intention through an integrated model, echoing the information systems success framework in which information quality, system quality and service quality are the exogenous drivers of use and user satisfaction (DeLone & McLean, 2003).

These lines have rarely been integrated. Where quality constructs are included in UTAUT-based specifications, they are usually entered as additional direct predictors of intention, placing them in competition with performance expectancy and effort expectancy for the same variance. Theoretically, this is unsatisfying: the DeLone and McLean (2003) formulation treats information quality as an antecedent of the beliefs that in turn drive use, not as a parallel belief. The present study follows that logic and positions perceived content quality upstream of effort expectancy, performance expectancy, trust, and self-regulated learning.

The trust–risk literature supplies the second integrative element. McKnight, Choudhury and Kacmar (2002) established trusting beliefs as a distinct construct with measurable antecedents, and Pavlou (2003) demonstrated that trust and perceived risk jointly qualify the technology acceptance model in transactional settings, with trust reducing risk and risk reducing intention. Featherman and Pavlou (2003) disaggregated perceived risk into performance, financial, time, psychological, social, and privacy facets. In the MOOC context, the salient facets are performance risk (whether the content matches formal instruction), time risk (whether the investment is repaid), privacy risk and, distinctively, credential risk (whether certificates are recognized). To our knowledge, no model has tested whether the aggregate of these facets inhibits experienced professional users, while also including a properly specified trust antecedent.

Finally, self-regulated learning is central to any account of MOOC engagement, because the format removes the external scaffolding of scheduled instruction. Zimmerman (2002) characterizes self-regulated learners as those who set goals, monitor progress, manage time, and adapt strategies. Environmental structuring is one mechanism that supports self-regulation, suggesting that well-organized, well-sequenced course content should raise rather than merely accompany self-regulatory behavior. This provides the rationale for modeling perceived content quality as an antecedent of self-regulated learning rather than a correlate.

1.4 State Hypotheses and Their Correspondence to Research Design

The hypothesized model is organized in four conceptual layers. The exogenous layer comprises facilitating conditions and social influence, which in UTAUT represent the resource and normative context of adoption. The platform-evaluation layer comprises perceived content quality and trust. The competence layer comprises digital–AI literacy and self-regulated learning. The belief layer comprises effort expectancy, performance expectancy, and perceived risk, and the outcome layer comprises behavioral intention and use behavior.

The central hypotheses are as follows. Perceived content quality positively influences effort expectancy, trust, and self-regulated learning because well-structured, academically credible material is easier to navigate, more deserving of confidence, and more supportive of independent study (DeLone & McLean, 2003; Zimmerman, 2002). Effort expectancy positively influences performance expectancy, reproducing the core mediating pathway of the technology acceptance model (Davis, 1989). Trust positively influences behavioral intention and negatively influences perceived risk (McKnight et al., 2002; Pavlou, 2003). Performance expectancy, social influence, and behavioral intention influence intention and use in the directions specified by UTAUT (Venkatesh et al., 2003). Following the dominant tradition, perceived risk is hypothesized to reduce behavioral intention and use behavior; this hypothesis is stated in its conventional directional form so that a contrary result is interpretable rather than post hoc.

The remaining structural relationships in the estimated model, including the paths from facilitating conditions and perceived content quality to digital–AI literacy and the direct paths from social influence to the competence and belief constructs, are exploratory. We report them in full with their significance levels, and we critically examine their directional plausibility in the Discussion. The cross-sectional survey design permits estimation of the specified associations and of their indirect transmission, but not the adjudication of causal direction; this constraint is stated explicitly and revisited in Section 4.6.

2. Method

2.1 Participants and Procedure

We collected data through a self-administered online questionnaire distributed to educational personnel in Thailand with prior experience using MOOC platforms. Participation was voluntary and anonymous, and respondents were informed of the study's purpose and their right to withdraw before providing consent. We received 290 completed questionnaires. After screening for insufficient-effort responding, as described in Section 2.3, the analytical sample comprised 270 respondents. This sample size satisfies conventional requirements for partial least squares structural equation modeling, exceeding ten times the largest number of structural paths directed at any single construct by a wide margin (Hair, Risher, Sarstedt, & Ringle, 2019).

2.2 Measures

All constructs were measured reflectively using multi-item scales adapted from established instruments and rated on a seven-point Likert scale anchored at 1 (strongly disagree) and 7 (strongly agree). We adapted performance expectancy, effort expectancy, social influence, facilitating conditions, behavioral intention, and use behavior from Venkatesh et al. (2003, 2012). Trust items were adapted from McKnight et al. (2002) and covered confidence in data privacy, the professional value of certificates, instruction quality, and platform stability. Perceived risk items were adapted from Featherman and Pavlou (2003). They covered concerns about content standards relative to formal education, return on time invested, personal data security, and recognition of certificates by institutions. Perceived content quality items reflected currency and professional relevance, academic accuracy, media quality and sequencing, consistent with the information quality dimension of DeLone and McLean (2003). Self-regulated learning items were adapted from Zimmerman (2002) and covered goal setting, time management, progress monitoring, self-motivation and strategy adaptation. Digital literacy and AI literacy items were developed from the four-aspect conceptualization of Ng et al. (2021), covering information search, digital content creation, digital communication, information evaluation, technical troubleshooting, understanding of AI and machine learning principles, applied use of AI tools, critical evaluation of AI limitations and bias, and awareness of AI ethics in education. The instrument was prepared in Thai, and subject-matter experts reviewed its content validity before administration.

2.3 Data Screening

We screened responses for insufficient-effort responding using a multi-index protocol (Curran, 2016). Intra-individual response variability (Dunn, Heggstad, Shanock, & Theilgard, 2018), maximum long string (Johnson, 2005), and Mahalanobis distance (Meade & Craig, 2012) were computed across all fifty indicators. Twenty respondents (6.9%) produced an intra-individual response variability of zero, that is, an identical

response to every item despite the presence of a negatively worded perceived-risk block; because such a pattern is logically inconsistent with the instrument design, these cases were removed, yielding an analytical sample of 270. Straightlining of this kind is a recognized threat to data quality in self-administered surveys and lacks a single standard detection technique, which motivated the use of several indices in combination (Kim, Dykema, Stevenson, Black, & Moberg, 2019; Marjanovic, Holden, Struthers, Cribbie, & Greenglass, 2015). Five missing values (SRL4, $n = 2$; SRL5, $n = 3$), representing 0.04% of all data points, were replaced with the corresponding item mean. A robustness check applying a stricter criterion, namely flagging on at least two of the three indices ($n = 244$), reproduced the same pattern of structural relationships.

2.4 Data Analysis

The model was estimated using partial least squares structural equation modeling in SmartPLS, employing the path weighting scheme. This method was selected because the model is complex, contains a comparatively large number of constructs and indicators relative to sample size, and is oriented towards prediction as well as explanation (Hair et al., 2019). The measurement model was assessed for indicator reliability, internal consistency, convergent validity, and discriminant validity, the latter using the heterotrait–monotrait ratio of correlations (Henseler, Ringle, & Sarstedt, 2015) alongside the Fornell and Larcker (1981) criterion. We tested structural relationships by bootstrapping with 10,000 subsamples, bias-corrected and accelerated confidence intervals, and two-tailed testing. Effect sizes were interpreted following Cohen (1988). We assessed out-of-sample predictive capability using PLSpredict with ten folds and ten repetitions (Shmueli et al., 2019) and the cross-validated predictive ability test (Liengaard et al., 2021; Sharma, Liengaard, Hair, Sarstedt, & Ringle, 2023). Common method variance was considered in the design through anonymity, item ordering, and the inclusion of reverse-direction content (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003), and assessed statistically through the full collinearity approach, in which each construct is regressed on all remaining constructs and the resulting variance inflation factors are inspected (Kock, 2015).

3. Results

3.1 Measurement Model

All fifty indicator loadings exceeded or approached the recommended threshold of .708, ranging from .701 (FC2) to .953 (PR2). FC2 was retained because its removal would not have raised composite reliability or average variance extracted for facilitating conditions above the required thresholds. Because digital literacy and AI literacy exhibited a heterotrait–monotrait ratio of .851 in a preliminary specification, exceeding the conservative .85 criterion (Henseler et al., 2015), the two indicator sets were combined into a single reflective construct designated digital–AI literacy. This decision was supported psychometrically: the nine-indicator construct was unidimensional, with a first-to-second eigenvalue ratio of 8.24; all loadings exceeded .80; average variance extracted reached .734; and Cronbach's alpha reached .955.

Table 1. Construct reliability and convergent validity

Construct	Cronbach's α	ρ_A	ρ_C	AVE
BI	.953	.953	.966	.877
DL&AIL	.955	.961	.961	.734
EE	.923	.927	.945	.811
FC	.812	.825	.877	.642
PCQ	.945	.947	.960	.857
PE	.913	.914	.939	.793
PR	.951	.965	.965	.872
SI	.909	.915	.936	.786
SRL	.929	.930	.946	.779
TR	.924	.929	.947	.816
UB	.958	.958	.969	.887

Note. Thresholds: $\alpha \geq .70$; $\rho_A \geq .70$; ρ_C between .70 and .95; AVE $\geq .50$.

As Table 1 shows, Cronbach's alpha ranged from .812 to .958, composite reliability ρ_A from .825 to .965, and ρ_C from .877 to .969, all exceeding the .70 criterion. Average variance extracted ranged from .642 to .887, confirming convergent validity for every construct.

Table 2. Discriminant validity: heterotrait–monotrait ratio of correlations

	BI	DL&AIL	EE	FC	PCQ	PE	PR	SI	SRL	TR
BI										
DL&AIL	.502									
EE	.471	.425								
FC	.492	.507	.487							
PCQ	.595	.450	.585	.594						
PE	.602	.322	.631	.510	.550					
PR	.236	.108	.109	.186	.101	.050				
SI	.475	.302	.512	.669	.490	.534	.259			
SRL	.663	.409	.536	.575	.821	.552	.174	.626		
TR	.682	.396	.542	.578	.803	.637	.062	.440	.796	
UB	.610	.442	.352	.407	.491	.265	.391	.408	.403	.456

Note. All values fall below the conservative .85 criterion; the highest is $PCQ \leftrightarrow SRL = .821$.

Discriminant validity was supported (Table 2). No ratio approached the liberal threshold of .90, and none exceeded the conservative threshold of .85. The three highest ratios involved conceptually adjacent evaluative constructs: perceived content quality with self-regulated learning (.821), perceived content quality with trust (.803), and self-regulated learning with trust (.796). All remaining ratios fell below .70.

Table 3. Discriminant validity: Fornell–Larcker criterion

	BI	DL&AIL	EE	FC	PCQ	PE	PR	SI	SRL	TR	UB
BI	.937										
DL&AIL	.483	.857									
EE	.446	.404	.901								
FC	.434	.458	.433	.801							
PCQ	.565	.431	.547	.529	.926						
PE	.563	.308	.582	.447	.510	.890					
PR	.229	.088	.100	.162	.092	-.031	.934				
SI	.446	.284	.474	.578	.456	.491	.246	.887			
SRL	.624	.388	.497	.506	.772	.511	.167	.581	.883		
TR	.639	.377	.506	.507	.755	.587	.034	.410	.741	.903	
UB	.583	.428	.333	.363	.466	.250	.379	.381	.381	.429	.942

Note. Diagonal elements (in the leading position of each row) are the square roots of the average variance extracted; off-diagonal elements are construct correlations.

The Fornell and Larcker (1981) criterion was likewise satisfied (Table 3). For every construct, the square root of the average variance extracted exceeded its correlation with all other constructs. The narrowest margins occurred for perceived content quality ($\sqrt{AVE} = .926$ against a maximum correlation of .772 with self-regulated learning) and trust ($\sqrt{AVE} = .903$ against .755 with perceived content quality), both comfortably above the criterion.

3.2 Structural Model

Collinearity was assessed at three levels. Inner-model variance inflation factors ranged from 1.000 ($FC \rightarrow PCQ$) to 2.974 ($SRL \rightarrow TR$), all below the conservative threshold of 3.3 recommended by Hair et al. (2019), indicating that the structural estimates are not distorted by predictor collinearity (Table 4).

Table 4. Inner-model collinearity statistics

Path	VIF	Path	VIF
BI → UB	1.779	PCQ → TR	2.915
DL&AIL → EE	1.351	PE → BI	2.000
DL&AIL → PE	1.391	PR → BI	1.106
DL&AIL → TR	1.404	PR → UB	1.091
EE → BI	1.714	SI → BI	1.551
EE → PE	1.534	SI → DL&AIL	1.575
EE → PR	1.527	SI → EE	1.577
EE → TR	1.636	SI → PE	1.888
EE → UB	1.503	SI → PR	1.639
FC → DL&AIL	1.732	SI → SRL	1.263
FC → EE	1.882	SI → TR	1.927
FC → PCQ	1.000	SI → UB	1.921
FC → PE	1.809	SRL → PE	1.782
FC → TR	1.884	SRL → PR	2.804
FC → UB	1.667	SRL → TR	2.974
PCQ → DL&AIL	1.457	SRL → UB	2.134
PCQ → EE	1.554	TR → BI	1.650
PCQ → SRL	1.263	TR → PR	2.378

Outer-model variance inflation factors ranged from 1.325 (FC2) to 6.294 (PR2), with twelve of the fifty indicators exceeding 5.0. Because all constructs in this model are specified reflectively, high indicator collinearity is expected rather than problematic: reflective indicators are conceived as interchangeable manifestations of the same latent variable, so elevated intercorrelation reflects a well-functioning scale. The variance inflation factor threshold of 5.0 applies to formatively measured constructs, where indicator redundancy would destabilize the weights, and is therefore not a validity criterion here (Hair et al., 2019).

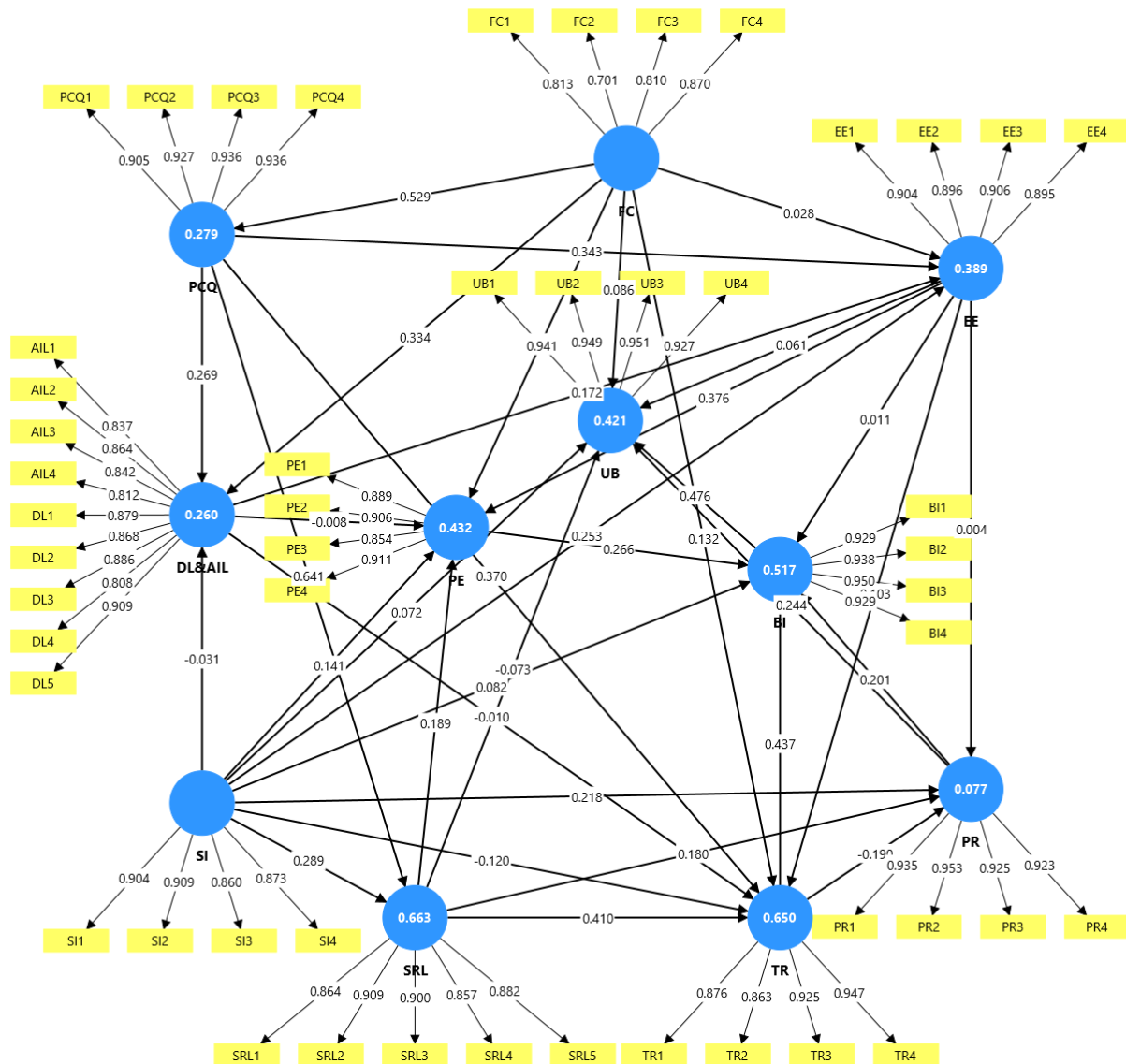
Common method variance was assessed statistically using the full collinearity approach (Kock, 2015), in which each construct is regressed on all remaining constructs. Full collinearity variance inflation factors ranged from 1.317 (PR) to 3.740 (SRL), as shown in Table 7. Three constructs, self-regulated learning (3.740), perceived content quality (3.378), and trust (3.358), marginally exceeded the conservative 3.3 criterion, consistent with their substantive conceptual proximity documented in the discriminant validity analysis. However, all values remained well below the 5.0 threshold at which common method bias is regarded as a serious threat. Combined with the procedural remedies described in Section 2.4, this indicates that common method variance is unlikely to account for the observed relationships (Podsakoff et al., 2003).

Table 5. Full collinearity variance inflation factors

Construct	VIF	Construct	VIF
BI	2.733	PR	1.317
DL&AIL	1.576	SI	2.202
EE	1.851	SRL	3.740
FC	2.006	TR	3.358
PCQ	3.378	UB	2.000
PE	2.247		

Note. Computed following Kock (2015). Values below 3.3 indicate no evidence of common method bias; values below 5.0 are generally regarded as acceptable.

Figure 1 presents the estimated model with standardized path coefficients and coefficients of determination, and Figure 2 presents the corresponding p-values.



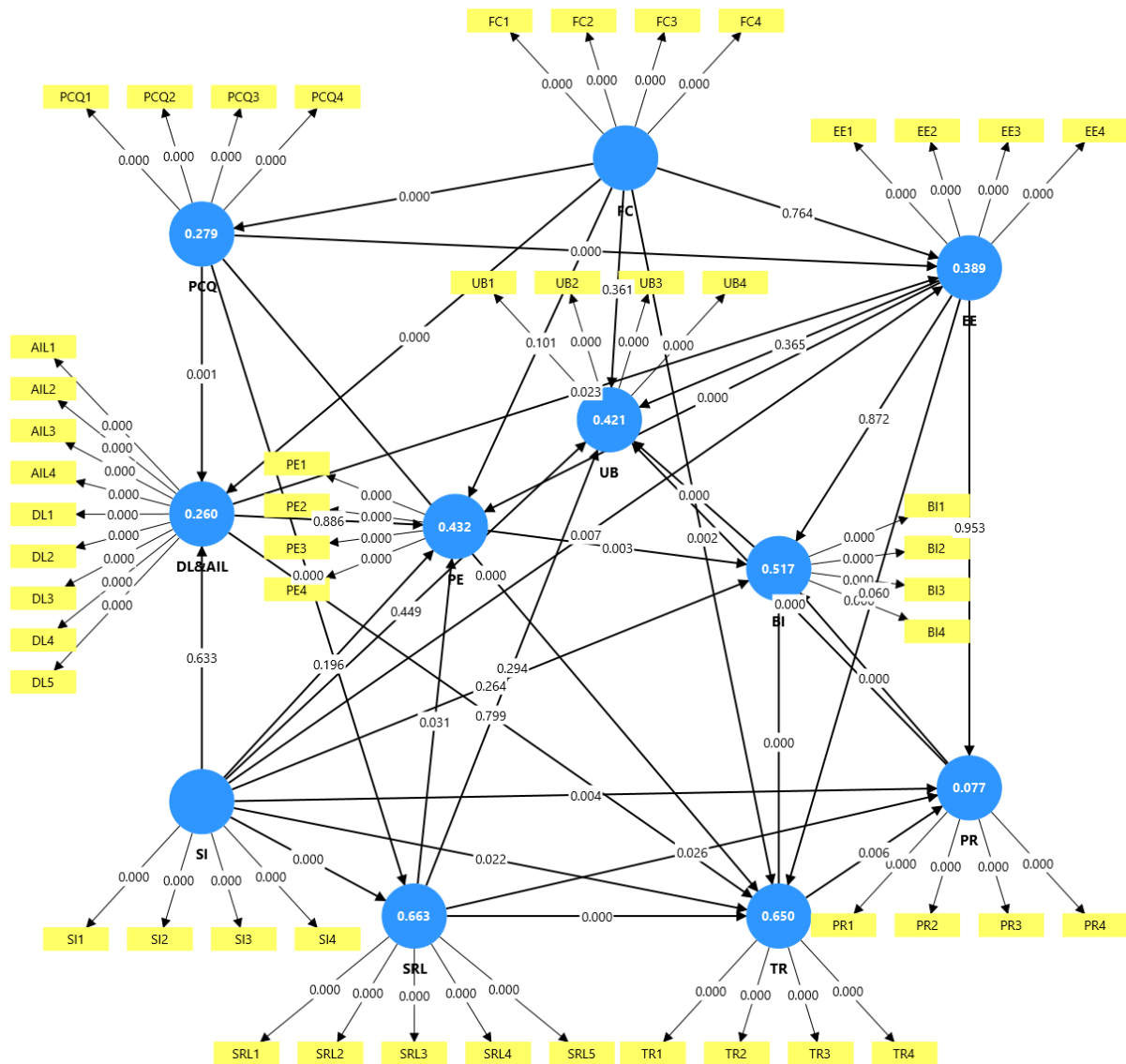


Table 6. Structural model path coefficients, significance testing and effect sizes

	Path	β	t	p	95% BCa CI	f ²	Result
H1	BI → UB	.476	7.751	< .001	[.358, .601]	.220	Supported
H2	DL&AIL → EE	.172	2.270	.023	[.053, .350]	.036	Supported
H3	DL&AIL → PE	-.008	0.144	.886	[-.123, .105]	.000	Not supported
H4	DL&AIL → TR	-.010	0.254	.799	[-.078, .073]	.000	Not supported
H5	EE → BI	.011	0.161	.872	[-.104, .162]	.000	Not supported
H6	EE → PE	.376	4.981	< .001	[.216, .511]	.162	Supported
H7	EE → PR	.004	0.059	.953	[-.128, .136]	.000	Not supported
H8	EE → TR	.103	1.884	.060	[-.002, .215]	.019	Not supported
H9	EE → UB	.061	0.906	.365	[-.075, .189]	.004	Not supported
H10	FC → DL&AIL	.334	4.044	< .001	[.193, .514]	.087	Supported
H11	FC → EE	.028	0.300	.764	[-.130, .229]	.001	Not supported
H12	FC → PCQ	.529	7.228	< .001	[.388, .671]	.388	Supported
H13	FC → PE	.111	1.639	.101	[-.010, .257]	.012	Not supported
H14	FC → TR	.132	3.060	.002	[.045, .217]	.027	Supported
H15	FC → UB	.086	0.913	.361	[-.120, .251]	.008	Not supported
H16	PCQ → DL&AIL	.269	3.208	.001	[.099, .427]	.067	Supported
H17	PCQ → EE	.343	4.068	< .001	[.167, .497]	.124	Supported
H18	PCQ → SRL	.641	16.185	< .001	[.566, .721]	.964	Supported
H19	PCQ → TR	.370	3.888	< .001	[.174, .550]	.135	Supported
H20	PE → BI	.266	2.944	.003	[.069, .422]	.073	Supported
H21	PR → BI	.201	5.444	< .001	[.127, .272]	.076	Supported
H22	PR → UB	.244	4.039	< .001	[.126, .364]	.095	Supported
H23	SI → BI	.082	1.117	.264	[-.056, .231]	.009	Not supported
H24	SI → DL&AIL	-.031	0.478	.633	[-.153, .103]	.001	Not supported
H25	SI → EE	.253	2.721	.007	[.056, .423]	.066	Supported
H26	SI → PE	.141	1.294	.196	[-.076, .349]	.018	Not supported
H27	SI → PR	.218	2.870	.004	[.072, .369]	.031	Supported
H28	SI → SRL	.289	6.124	< .001	[.192, .375]	.195	Supported
H29	SI → TR	-.120	2.296	.022	[-.219, -.015]	.021	Supported
H30	SI → UB	.072	0.757	.449	[-.098, .278]	.005	Not supported
H31	SRL → PE	.189	2.155	.031	[.026, .367]	.035	Supported
H32	SRL → PR	.180	2.225	.026	[.015, .330]	.013	Supported
H33	SRL → TR	.410	4.532	< .001	[.227, .585]	.162	Supported
H34	SRL → UB	-.073	1.050	.294	[-.208, .065]	.004	Not supported
H35	TR → BI	.437	6.944	< .001	[.309, .560]	.239	Supported
H36	TR → PR	-.190	2.744	.006	[-.325, -.054]	.016	Supported

Note. β = standardised path coefficient; BCa CI = bias-corrected and accelerated bootstrap confidence interval (10,000 subsamples); f² interpreted as .02 small, .15 medium, .35 large (Cohen, 1988). "Supported" denotes statistical significance at $p < .05$ and does not imply that the observed sign matches the hypothesized direction; see Section 4.3 regarding perceived risk.

Table 7. Coefficients of determination for the endogenous constructs

Endogenous construct	R ²	t	p
BI	.517	10.036	< .001
DL&AIL	.260	2.660	.008
EE	.389	6.938	< .001
PCQ	.279	3.597	< .001
PE	.432	7.826	< .001
PR	.077	2.609	.009
SRL	.663	15.231	< .001
TR	.650	16.997	< .001
UB	.421	8.937	< .001

The model explained a substantial proportion of variance in the principal endogenous constructs (Table 8). Self-regulated learning showed the highest explanatory power ($R^2 = .663$), followed by trust (.650), behavioral intention (.517), performance expectancy (.432), use behavior (.421), and effort expectancy (.389). Perceived risk was the least well explained construct (.077), although the coefficient remained significant ($p = .009$). All coefficients of determination were significant at $p < .01$.

Twenty-one of the thirty-six structural relationships were statistically significant. Behavioral intention was the dominant proximal determinant of use behavior ($\beta = .476$, $t = 7.75$, $p < .001$, $f^2 = .220$), and trust was the strongest predictor of behavioral intention ($\beta = .437$, $t = 6.94$, $p < .001$, $f^2 = .239$). Perceived content quality exerted a very large effect on self-regulated learning ($\beta = .641$, $t = 16.18$, $p < .001$, $f^2 = .964$) and substantial effects on trust ($\beta = .370$, $p < .001$) and effort expectancy ($\beta = .343$, $p < .001$). The technology acceptance pathway from effort expectancy to performance expectancy was strongly supported ($\beta = .376$, $t = 4.98$, $p < .001$, $f^2 = .162$), and performance expectancy, in turn, predicted behavioral intention ($\beta = .266$, $t = 2.94$, $p = .003$).

Two results depart from conventional expectations. Trust significantly reduced perceived risk ($\beta = -.190$, $t = 2.74$, $p = .006$), consistent with the trust–risk framework (Pavlou, 2003). However, perceived risk exerted a significant positive effect on both behavioral intention ($\beta = .201$, $t = 5.44$, $p < .001$, 95% CI [.127, .272]) and use behavior ($\beta = .244$, $t = 4.04$, $p < .001$, 95% CI [.126, .364]). Both confidence intervals excluded zero by a considerable margin. Social influence displayed a negative association with trust ($\beta = -.120$, $t = 2.30$, $p = .022$).

Several relationships were not supported. Effort expectancy did not directly predict behavioral intention ($\beta = .011$, $p = .872$), with its influence operating entirely through performance expectancy. Digital–AI literacy showed no significant direct effect on performance expectancy ($p = .886$) or trust ($p = .799$). The direct paths from social influence to behavioral intention ($p = .264$) and from facilitating conditions to use behavior ($p = .361$) were not significant in the presence of the extended predictor set.

3.3 Mediation Analysis

Forty-five specific indirect effects were statistically significant; Table 8 reports the twelve largest.

Table 8. Significant specific indirect effects (twelve largest)

Indirect path	β	t	p
FC → PCQ → SRL	.339	6.822	< .001
PCQ → SRL → TR	.263	4.411	< .001
TR → BI → UB	.208	4.545	< .001
FC → PCQ → TR	.196	3.393	< .001
FC → PCQ → EE	.181	3.746	< .001
PCQ → TR → BI	.162	3.160	.002
FC → PCQ → DL&AIL	.142	3.211	.001
FC → PCQ → SRL → TR	.139	3.720	< .001
PE → BI → UB	.127	3.097	.002
PCQ → SRL → PE	.121	2.062	.039
PCQ → SRL → PR	.115	2.185	.029
PCQ → SRL → TR → BI	.115	3.611	< .001

The mediation results clarify the model's architecture. Facilitating conditions reached self-regulated learning entirely through perceived content quality ($\beta = .339$, $p < .001$), and perceived content quality reached trust both directly and through self-regulated learning ($\beta = .263$, $p < .001$). Trust influenced use behavior through behavioral intention ($\beta = .208$, $p < .001$). The sequence from effort expectancy through performance expectancy to behavioral intention was significant ($\beta = .100$, $p = .004$), confirming that effort expectancy operates indirectly. Perceived risk carried a significant positive indirect effect on use behavior through behavioral intention ($\beta = .096$, $p < .001$).

Table 9. Total effects on behavioral intention and use behavior

Path	β	t	p
BI $\rightarrow$ UB	.476	7.751	< .001
DL&AIL $\rightarrow$ BI	.020	0.629	.529
DL&AIL $\rightarrow$ UB	.020	0.863	.388
EE $\rightarrow$ BI	.153	2.755	.006
EE $\rightarrow$ UB	.130	1.657	.097
FC $\rightarrow$ BI	.287	5.612	< .001
FC $\rightarrow$ UB	.207	2.286	.022
PCQ $\rightarrow$ BI	.366	7.042	< .001
PCQ $\rightarrow$ UB	.149	3.063	.002
PE $\rightarrow$ BI	.266	2.944	.003
PE $\rightarrow$ UB	.127	3.097	.002
PR $\rightarrow$ BI	.201	5.444	< .001
PR $\rightarrow$ UB	.340	5.975	< .001
SI $\rightarrow$ BI	.226	2.705	.007
SI $\rightarrow$ UB	.239	2.706	.007
SRL $\rightarrow$ BI	.250	3.777	< .001
SRL $\rightarrow$ UB	.071	0.941	.347
TR $\rightarrow$ BI	.398	5.881	< .001
TR $\rightarrow$ UB	.143	2.508	.012

When direct and indirect effects are combined (Table 9), trust (total $\beta = .398$, $p < .001$) and perceived content quality ($\beta = .366$, $p < .001$) emerge as the most influential antecedents of behavioral intention, followed by facilitating conditions ($\beta = .287$, $p < .001$) and self-regulated learning ($\beta = .250$, $p < .001$). For use behavior, behavioral intention remained dominant ($\beta = .476$), followed by perceived risk ($\beta = .340$, $p < .001$), social influence ($\beta = .239$, $p = .007$) and facilitating conditions ($\beta = .207$, $p = .022$). Social influence and facilitating conditions, whose direct effects were non-significant, therefore exert significant total effects, indicating that their influence is transmitted through the platform-evaluation and belief constructs rather than operating directly.

3.4 Predictive Assessment

Table 10. PLSpredict summary at the construct level

Construct	Q ² predict	RMSE	MAE
BI	.222	0.894	0.605
DL&AIL	.176	0.929	0.606
EE	.232	0.885	0.659
PCQ	.258	0.869	0.618
PE	.253	0.876	0.644
PR	.047	0.983	0.846
SRL	.354	0.815	0.604
TR	.243	0.878	0.642
UB	.154	0.927	0.686

Table 11. Indicator-level comparison of PLS-SEM against the linear model benchmark

Construct	Indicators with PLS-SEM RMSE < LM RMSE
AIL	2 / 4
BI	2 / 4
DL	1 / 5
EE	0 / 4
PCQ	0 / 4
PE	1 / 4
PR	0 / 4
SRL	1 / 5
TR	0 / 4
UB	2 / 4

All Q^2_{predict} values were positive, ranging from .047 to .354, indicating that the model outperforms the naive indicator-average benchmark. At the indicator level, however, the PLS-SEM root mean squared error was lower than the linear model benchmark for only nine of the forty-two indicators. Following the guidelines of Shmueli et al. (2019), this pattern indicates low predictive power, and we report it rather than inferring favorable results from the positive Q^2_{predict} values alone.

Table 12. Cross-validated predictive ability test

Construct	vs IA: Δloss	t	p	vs LM: Δloss	t	p
BI	-.237	3.883	< .001	.011	0.291	.771
DL&AIL	-.100	2.121	.035	.016	0.837	.403
EE	-.169	3.386	< .001	.085	2.323	.021
PCQ	-.153	2.528	.012	.015	0.437	.662
PE	-.175	3.353	< .001	.028	1.210	.227
PR	-.107	1.432	.153	.051	0.699	.485
SRL	-.207	3.738	< .001	.014	0.621	.535
TR	-.185	3.322	.001	.064	1.639	.102
UB	-.285	3.501	< .001	.004	0.072	.943
Overall	-.171	4.094	< .001	.030	1.422	.156

Note. IA = indicator-average benchmark; LM = linear model benchmark. Negative values indicate that PLS-SEM outperforms the benchmark.

The cross-validated predictive ability test yields a more favorable assessment. Against the indicator-average benchmark, the average loss difference was negative and significant for every construct except perceived risk, with an overall value of $-.171$ ($t = 4.09$, $p < .001$), confirming predictive validity. Against the linear model benchmark, the overall average loss difference was $.030$ ($t = 1.42$, $p = .156$), not significantly different from zero, indicating that the theoretically derived model performs no worse than an atheoretical linear specification. Only effort expectancy showed a significant disadvantage relative to the linear model ($\Delta\text{loss} = .085$, $p = .021$).

3.5 Model Fit

The standardized root mean square residual was $.061$ for the saturated model and $.085$ for the estimated model. Consistent with current methodological guidance, fit indices in partial least squares structural equation modeling remain contested and are reported in Table 13, here only as approximate indicators; we do not report the normed fit index because no generally accepted cut-off exists within this framework (Hair et al., 2019). Model quality is instead evaluated based on explanatory power, effect size, and the out-of-sample predictive assessment reported in Section 3.4.

Table 13. Model Fit

	Saturated model	Estimated model
SRMR	0.061	0.085
d_ULS	4.771	9.316
d_G	4.038	4.265
Chi-square	5256.966	5265.343
NFI	0.702	0.701

4. Discussion

4.1 Perceived Content Quality as the Upstream Driver

The most consistent finding of this study is the upstream position of perceived content quality. It had the largest single effect on self-regulated learning and also shaped trust and effort expectancy; its total effect on behavioral intention (.366) was second only to trust. This pattern supports the information systems success logic in which information quality is an antecedent of the beliefs that drive use rather than a belief in its own right (DeLone & McLean, 2003), and it extends prior MOOC quality research (Gu et al., 2021) by specifying the mechanisms of transmission. The large coefficient linking perceived content quality to self-regulated learning also gives empirical substance to Zimmerman's (2002) notion of environmental structuring: educational personnel who judge course material as accurate, current, and well-sequenced report correspondingly stronger goal setting, monitoring, and strategy adaptation.

4.2 Trust and the Trust–Risk Mechanism

Trust was the strongest direct predictor of behavioral intention and the largest single contributor to explained variance in intention. This finding positions trust, rather than performance expectancy, as the central evaluative construct for professional MOOC users, consistent with the argument that when the value of an offering cannot be verified in advance, trusting beliefs substitute for direct evidence (McKnight et al., 2002). The significant negative path from trust to perceived risk reproduces the trust–risk mechanism reported in transactional contexts (Pavlou, 2003) and indicates that it generalizes to professional learning platforms.

A secondary result deserves comment. Social influence was negatively associated with trust. When institutional policy and colleagues encourage adoption, respondents reported lower, not higher, confidence in the platform. One reading is that normative pressure and genuine confidence are distinct and, in mandated contexts, potentially opposed: educators who use MOOCs largely because they are expected to may have had less opportunity to develop trust through direct evaluation. This interpretation aligns with the observation that the direct path from social influence to behavioral intention was not significant, while its total effect was, indicating that normative pressure operates through evaluative constructs rather than directly on intention.

4.3 Informed Adoption: Reinterpreting Perceived Risk

The most theoretically consequential finding is that perceived risk positively predicted both behavioral intention and use behavior, contrary to the hypothesized direction and the dominant treatment of risk as an inhibitor (Featherman & Pavlou, 2003; Pavlou, 2003). Three features of the result argue against dismissing it as an artifact. First, both coefficients were highly significant with confidence intervals well clear of zero. Second, the perceived-risk construct met every reliability and validity criterion, with loadings between .923 and .953. Third, perceived risk had a theoretically specified antecedent in trust, whose negative effect behaved exactly as predicted; the positive downstream association therefore cannot be attributed to the construct floating free of the nomological network.

We interpret the pattern as informed adoption. The respondents were educational personnel with existing MOOC experience, and the risk items concerned content standards relative to formal education, the return on time invested, data security, and the institutional recognition of certificates. These are not the concerns of a hesitant non-adopter; they are the concerns of someone who has used the platform enough to know its limitations. On this reading, risk perception among experienced professional users is partly a consequence of engagement rather than solely a barrier to it, and the conventional inhibitor formulation may be specific to pre-adoption or transactional settings. The practical implication is notable: institutions that respond to expressed concern about MOOC credentials by treating it as resistance to be overcome may be misreading a signal generated by the most engaged members of their workforce.

4.4 Theoretical Implications

Three implications follow. First, the complete mediation of effort expectancy by performance expectancy indicates that the classical technology acceptance pathway (Davis, 1989) remains operative within a UTAUT specification and that omitting it, as many UTAUT applications do, forces effort expectancy to compete directly with performance expectancy and suppresses its apparent contribution. Second, the results suggest that quality constructs should be positioned upstream of, not parallel to, the UTAUT belief constructs. Third, and most substantively, the informed-adoption pattern indicates that the sign of the risk–intention relationship may depend on the adoption stage and the user's professional standing, a boundary condition that acceptance theory does not currently articulate.

4.5 Practical Implications

For platform providers, the primacy of perceived content quality implies that investment in curation, sequencing, and academic accuracy is likely to yield greater returns than investment in interface refinement, since content quality influenced effort expectancy and trust. For institutions, the finding that facilitating conditions and social influence exert significant total but non-significant direct effects suggests that providing equipment and encouragement is necessary but not sufficient: their influence materializes only when the content itself is judged credible. For policy makers, the credential-recognition component of perceived risk points to a concrete lever. Formal articulation between MOOC certificates and professional advancement frameworks would address the specific concern most strongly voiced by experienced users.

4.6 Limitations and Future Research

Several limitations qualify these findings. The design is cross-sectional, so the structural coefficients should be read as associations consistent with the specified model rather than as established causal effects. This caution

applies especially to three specification decisions. The model treats facilitating conditions and perceived content quality as antecedents of digital–AI literacy, yet digital and AI literacy are individual competencies typically acquired prior to and independently of any specific platform, and a reversed or reciprocal specification is equally plausible. The path from facilitating conditions to perceived content quality links an individual-level resource condition to a platform attribute. It is more plausibly interpreted as reflecting a shared institutional context, in which better-resourced settings provide access to higher-quality offerings, than as a direct causal effect. The positive association between perceived risk and use behavior, discussed above, is likewise open to a reverse-causal reading in which use generates awareness rather than awareness generating use.

Second, all data are self-reported, including use behavior; platform-derived log data would provide a stronger behavioral criterion. Third, the indicator-level predictive assessment showed low predictive power, which limits confidence in applying the model to new samples, even though the cross-validated predictive ability test confirmed that the model performs no worse than a linear benchmark. Fourth, the merger of digital literacy and AI literacy, although psychometrically justified, forecloses examination of their distinct contributions; instruments designed to separate them more sharply would allow that question to be revisited. Fifth, the sample is drawn from a single national context, and the salience of credential recognition in particular is likely to vary with national qualification frameworks.

Future research should use longitudinal panel designs that can distinguish whether risk perception precedes or follows sustained use, and should test the informed-adoption interpretation directly by comparing novice and experienced users, following the moderating logic of experience established by Tseng et al. (2022). Experimental manipulation of platform-level attributes such as content sequencing and credential framing would allow causal testing of the upstream role of content quality.

5. Conclusion

This study developed and tested an extended UTAUT model of MOOC acceptance among educational personnel, incorporating perceived content quality, trust, perceived risk, self-regulated learning, and digital–AI literacy, and linking the dependent variable to self-reported use behavior. The model explained 51.7% of the variance in behavioral intention and 42.1% in use behavior. Perceived content quality emerged as the principal upstream driver and trust as the strongest determinant of intention, while effort expectancy operated entirely through performance expectancy. Contrary to the prevailing barrier view, perceived risk was positively associated with both intention and use, a pattern interpreted as informed adoption among experienced professional users. Taken together, the results suggest that acceptance models for professional learning platforms should treat content quality as an antecedent rather than a parallel belief, should model trust and risk as a linked pair, and should not assume that the sign of the risk relationship is invariant across adoption stages.

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Author contributions

Phanuwat Ruangkulsap: Conceptualization, methodology, investigation, data curation, formal analysis, visualization, and writing original draft.

Thada Jantakoon: Conceptualization, methodology, supervision, validation, and writing review and editing.
Rukthin Laoha: Validation, supervision, and writing review and editing

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Competing interests

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Informed consent

Informed consent was obtained from all participants prior to their participation in the study. Participation was voluntary and anonymous, and participants were informed of the purpose of the study and their right to withdraw before completing the online questionnaire.

Ethics approval

Participation was voluntary and anonymous, and no personally identifiable information was collected. The study involved a non-interventional, anonymous online questionnaire administered to educational personnel. Under the

applicable institutional requirements, formal ethics approval was not required for this study.

Provenance and peer review

Not commissioned; externally double-blind peer reviewed.

Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Data sharing statement

De-identified data supporting the findings of this study may be made available by the corresponding author upon reasonable request, subject to applicable institutional and data-protection requirements.

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Appendixes

Appendix A. Measurement Instrument

All items were rated on a seven-point Likert scale anchored at 1 (strongly disagree) and 7 (strongly agree). The instrument was administered in Thai; the English wording below is a translation prepared for reporting purposes. Loadings are the standardised outer loadings obtained in the final model (n = 270). Items DL1-DL5 and AIL1-AIL4 were combined into the single reflective construct digital-AI literacy (DL&AIL) following the

discriminant validity assessment reported in Section 3.1.

Table A1. Constructs, measurement items and standardised outer loadings

Code	Item	Loading
Performance expectancy (PE) — Venkatesh et al. (2003, 2012)		
PE1	Using MOOC platforms improves the effectiveness of my teaching and instructional management.	.889
PE2	MOOCs help me develop my professional skills more effectively.	.906
PE3	MOOCs enable me to access new knowledge rapidly.	.854
PE4	Using MOOCs helps me achieve my professional development goals more quickly.	.911
Effort expectancy (EE) — Venkatesh et al. (2003, 2012)		
EE1	Using MOOC platforms is easy for me.	.904
EE2	I can learn how to use MOOCs quickly.	.896
EE3	The interface of MOOC platforms is easy to understand and convenient to use.	.906
EE4	I can use the various features of MOOCs proficiently.	.895
Social influence (SI) — Venkatesh et al. (2003, 2012)		
SI1	Administrators and key personnel in my institution support my use of MOOCs.	.904
SI2	My institution's administrators encourage me to use MOOCs to improve my teaching.	.909
SI3	My colleagues use MOOCs and recommend that I use them as well.	.860
SI4	My institution has a policy promoting the use of MOOCs for staff competency development.	.873
Facilitating conditions (FC) — Venkatesh et al. (2003, 2012)		
FC1	I have the devices and resources necessary to use MOOCs (e.g., computer, smartphone).	.813
FC2	My institution provides technical support for using MOOCs.	.701
FC3	The internet connection I use is stable and sufficient for using MOOCs.	.810
FC4	MOOC platforms are compatible with the other tools and systems I use in my work.	.870
Digital literacy (DL) — Developed from Ng et al. (2021)		
DL1	I can use digital tools to search for and compile information effectively.	.879
DL2	I can create and edit digital content in a variety of formats (e.g., documents, presentations, video).	.868
DL3	I can use digital technology to communicate and collaborate with others effectively.	.886
DL4	I can evaluate the credibility and accuracy of information found in online media.	.808
DL5	I can independently solve basic problems relating to digital technology.	.909
AI literacy (AIL) — Developed from Ng et al. (2021)		
AIL1	I understand the fundamental principles of artificial intelligence and machine learning.	.837
AIL2	I can apply AI tools (e.g., ChatGPT or content recommendation systems) to support learning and teaching.	.864
AIL3	I can analyse and evaluate the limitations and potential biases of AI systems.	.842
AIL4	I am aware of the ethical issues associated with the use of AI in educational contexts.	.812
Trust (TR) — McKnight et al. (2002)		
TR1	I am confident that MOOC platforms protect the privacy of my personal data.	.876
TR2	I am confident that certificates obtained from MOOCs are valuable and professionally recognised.	.863
TR3	I am confident that MOOCs provide high-quality and reliable learning services.	.925
TR4	I am confident in the stability and security of MOOC platform systems.	.947
Perceived content quality (PCQ) — Developed from DeLone and McLean (2003)		
PCQ1	The content on MOOC platforms is current and responsive to my professional needs.	.905
PCQ2	MOOC content is accurate, complete and academically credible.	.927
PCQ3	MOOC learning materials (e.g., videos, documents, quizzes) are of high quality and suitable for application.	.936
PCQ4	MOOC content is systematically sequenced and structured, enabling continuous and efficient learning.	.936
Self-regulated learning (SRL) — Adapted from Zimmerman (2002)		
SRL1	I can set my own learning goals when using MOOC platforms.	.864
SRL2	I can manage my time for learning through MOOCs effectively.	.909
SRL3	I can monitor and evaluate my own learning progress consistently.	.900
SRL4	I am motivated to learn through MOOCs on my own, even without an instructor supervising me.	.857
SRL5	I can adapt my learning strategies appropriately when I encounter difficulties in learning through MOOCs.	.882
Perceived risk (PR) — Featherman and Pavlou (2003)		
PR1	I am concerned that MOOC content may not meet the same standards as formal education.	.935
PR2	I am concerned that the time invested in learning through MOOCs may not be worthwhile relative to the outcomes.	.953
PR3	I am concerned about the security of my personal data on MOOC platforms.	.925
PR4	I am concerned that MOOC certificates may not be recognised by educational institutions or relevant agencies.	.923
Behavioural intention (BI) — Venkatesh et al. (2003, 2012)		
BI1	I intend to continue using MOOCs in the future.	.929
BI2	I plan to make MOOCs part of my instructional and professional development.	.938
BI3	I will recommend MOOCs to colleagues and other educational personnel.	.950
BI4	I expect to use MOOCs more frequently over the next 12 months.	.929
Use behaviour (UB) — Venkatesh et al. (2003, 2012)		
UB1	I regularly use MOOCs for learning or integrate them into my teaching.	.941
UB2	I use MOOCs to develop my professional skills on a continuing basis.	.949

UB3	I have used MOOCs consistently over the past six months.	.951
UB4	I use multiple MOOC platforms for learning and professional development.	.927

Note. Rows in bold type identify the construct and the source of its items. All loadings are significant at $p < .001$.

Appendix B. Summary of Constructs and Variables

Table B1. Constructs, their role in the model, number of indicators and explained variance

Code	Construct	Role in the model	Items	Mode	R ²
PE	Performance expectancy	Endogenous belief	4	Reflective	.432
EE	Effort expectancy	Endogenous belief	4	Reflective	.389
SI	Social influence	Exogenous	4	Reflective	—
FC	Facilitating conditions	Exogenous	4	Reflective	—
DL&AIL	Digital–AI literacy	Endogenous competence	9	Reflective	.260
TR	Trust	Endogenous evaluation	4	Reflective	.650
PCQ	Perceived content quality	Endogenous evaluation	4	Reflective	.279
SRL	Self-regulated learning	Endogenous competence	5	Reflective	.663
PR	Perceived risk	Endogenous belief	4	Reflective	.077
BI	Behavioural intention	Endogenous outcome	4	Reflective	.517
UB	Use behaviour	Endogenous outcome	4	Reflective	.421

Note. Social influence and facilitating conditions are the only fully exogenous constructs in the estimated model; a dash denotes that no R² is estimated for them. All other constructs are endogenous. Total indicators = 50.